1-HOUR HANDS-ON WORKSHOP

R Basics for Data Science & Analytics

Master core data structures, `dplyr` wrangling, basic statistics, and `ggplot2` visualizations.

Hands-On Code Exercises

Tidyverse Ecosystem

WORKSHOP BLUEPRINT

Agenda & Learning Path

01

Foundations

Vectors, Data Frames, Factors, and Lists.

02

Wrangling

Pipelines with `dplyr` verbs and native pipe ▷.

03

Analysis

Summary statistics, cross-tabs, and linear models.

04

Visualization

Layered charts using `ggplot2` grammar.

MODULE 01 · FOUNDATIONS

Core Data Containers

Vector `c()`

Atomic 1D sequence of uniform data types.

Data Frame `data.frame()`

2D tabular grid with mixed column types.

Factor `factor()`

Categorical vector with defined levels.

List `list()`

Flexible object holding heterogenous types.

HANDS-ON PRACTICE 01

Exercise: Building Data Frames

```r
# 1. Create vectors and assemble data frame
names  <- c("Ana", "Bob", "Cara", "Dan")
salary <- c(65000, 72000, 58000, 81000)
dept   <- factor(c("IT", "HR", "IT", "Finance"))

# Create tabular dataset
employees <- data.frame(names, salary, dept)

# Inspect structure
str(employees)
dim(employees)
```

Your Task

- Run script in RStudio console.
- Verify vector lengths match.
- Check dimensions with `dim()`.

MODULE 02 · WRANGLING

The `dplyr` Pipeline Verbs

`filter()`

Subset Rows

Extract observations meeting logical conditions.

`select()`

Pick Columns

Keep or rename specific variables.

`mutate()`

Create Fields

Compute new transformed variables.

`summarize()`

Aggregate Data

Collapse groups into statistical metrics.

HANDS-ON PRACTICE 02

Exercise: Data Wrangling Pipeline

```r
# Load dplyr and build pipe pipeline
library(dplyr)

dept_summary <- employees |>
  filter(salary > 60000) |>
  mutate(monthly = salary / 12) |>
  group_by(dept) |>
  summarize(
    avg_salary = mean(salary),
    count      = n()
  )
```

Pipeline Steps

- Use native pipe ▷ to chain commands.
- Filter salaries above 60k threshold.
- Compute group averages per department.

MODULE 03 · ANALYSIS

Descriptive & Statistical Functions

Function	Category	Primary Use Case
`mean(), median(), sd()`	Summary Statistics	Central tendency and dispersion measures.
`table(), prop.table()`	Frequency Tables	Categorical counts and relative proportions.
`cor()`	Correlation	Linear relationship strength between numeric variables.
`lm()`	Linear Regression	Fitting linear models (y ~ x).

HANDS-ON PRACTICE 03

Exercise: Statistical Summaries & Modeling

```r
# Run summary stats and linear regression

# Frequency distribution
table(employees$dept)

# Summary statistics
summary(employees$salary)

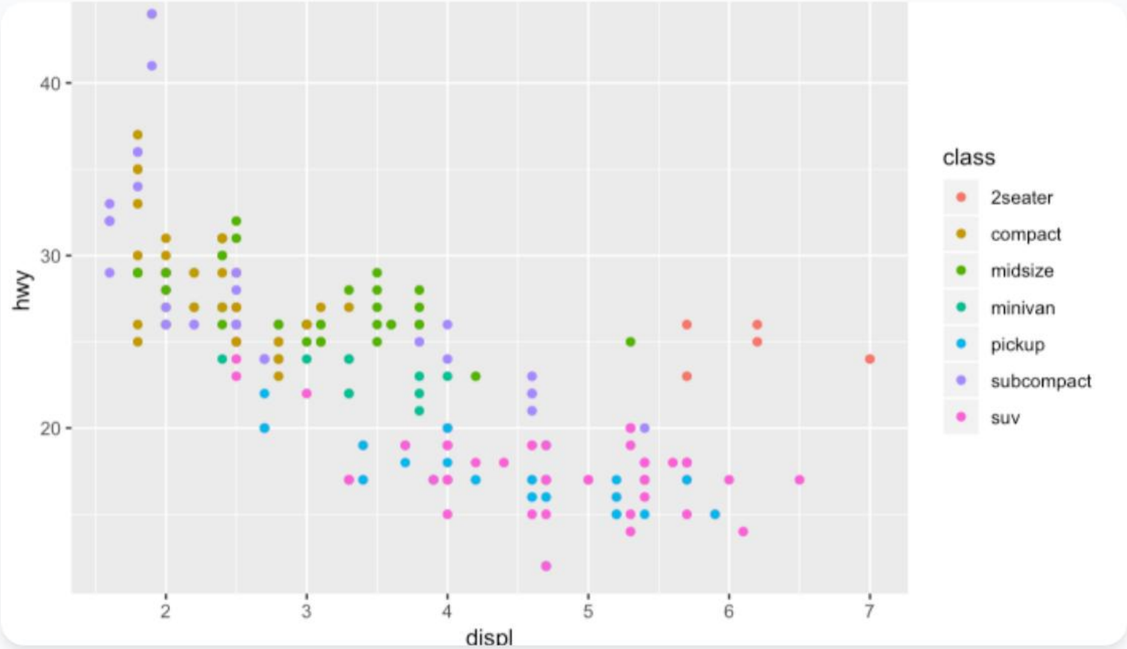
# Fit linear model
model <- lm(salary ~ dept, data = employees)
summary(model)
```

Analytical Goals

- Examine 5-number summary values.
- Check categorical frequency table.
- Evaluate model summary coefficients.

MODULE 04 · VISUALIZATION

Grammar of Graphics (`ggplot2`)



1. Data Layer

Clean input data frame passed to `ggplot()`.

2. Aesthetic Mapping `aes()`

Map variables to x/y axes, color, fill, or size.

3. Geometries `geom_*()`

Visual shapes: points, bars, boxplots, or lines.

HANDS-ON PRACTICE 04

Exercise: Layered Plot Creation

```
# Build layered bar chart in ggplot2
library(ggplot2)

ggplot(employees, aes(x = dept, y = salary, fill = dept)) +
  geom_col() +
  labs(
    title = "Salary Distribution by Department",
    x = "Department",
    y = "Salary ($)"
  ) +
  theme_minimal()
```

Plot Breakdown

- Map x and y variables in `aes()`.
- Add `geom_col()` column layer.
- Add labels and clean minimalist theme.

PACKAGE CHEAT SHEET

Core Tidyverse Packages Summary

readr / readxl

Fast data import for CSV and Excel files into clean data frames.

dplyr

Grammar for intuitive data manipulation and aggregation.

tidyr

Reshaping data between wide and long formats (`pivot_*`).

ggplot2

Declarative graphics framework based on layered aesthetics.

WORKSHOP WRAP-UP

Key Takeaways & Next Steps

- Store observations in structured **Data Frames**.
- Use native pipe `|>` for clean, readable pipelines.
- Build graphics incrementally using `ggplot2` layers.

Practice datasets to explore: `palmerpenguins`, `nycflights13`, `mtcars`

Image Sources



https://images.ctfassets.net/fi0zmnwlsnja/1cBXRR6naLZxfJLVrxQMUm/db1a35a0d9f7cff3f28f6a0c6de54c56/r-data-viz-01.png

Source: mode.com